

☒ Easy Level (10 Questions)

Q1. How do you define a tuple with one element?

- A. (1)
- B. (1,)
- C. [1]
- D. {1}

Answer: B

Explanation: Single-element tuples need a trailing comma.

Q2. Which of the following is immutable?

- A. List
- B. Tuple
- C. Dictionary
- D. Set

Answer: B

Explanation: Tuples cannot be changed once created.

Q3. How do you assign the same value to multiple variables?

- A. x = y = z = 5
- B. x, y, z = 5
- C. x = 5; y; z

D. None

Answer: A

Explanation: Chained assignment sets all to 5.

Q4. What does this print?

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```
name = "Komal" print(f"Hello {name}")
```

A. Hello name

B. Hello Komal

C. fHello Komal

D. Error

Answer: B

Explanation: f-string substitutes variable values.

Q5. How do you join two lists a and b?

- A. a + b
- B. a.append(b)
- C. a.extend(b)
- D. Both A and C

Answer: D

Explanation: Both add elements of b to a (extend modifies a in place).

Q6. Which method splits a string into a list?

- A. split()
- B. join()
- C. slice()
- D. divide()

Answer: A

Explanation: split() divides a string based on delimiter.

Q7. What is printed by this code?

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a, b = 5, 10 print(a, b)

- A. 5 10
- B. (5, 10)
- C. a b
- D. 10 5

Answer: A

Explanation: Tuple unpacking assigns values accordingly.

Q8. How do you declare a variable in Python?

- A. var x = 5
- B. int x = 5
- C. x = 5
- D. declare x = 5

Answer: C

Explanation: Python variables are assigned directly.

Q9. Which of the following formats strings using % operator?

- A. "Name: %s" % "Komal"
- B. "Name: {}".format("Komal")
- C. f"Name: Komal"
- D. Name % Komal

Answer: A

Explanation: % operator is old-style formatting.

Q10. What is output of:

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t = (1, 2, 3) print(type(t))

A. <class 'tuple'>

B. <class 'list'>

C. <class 'dict'>

D. <class 'int'>

Answer: A

Explanation: t is a tuple.

☒ Medium Level (20 Questions)

Q11. What happens if you try to change a tuple element?

- A. Changes successfully
- B. Error: tuples are immutable
- C. Changes only first element
- D. Creates new tuple automatically

Answer: B

Explanation: Tuples cannot be modified.

Q12. Which of the following is correct for multiple assignment?

- A. `x, y = 1, 2`
- B. `(x, y) = (1, 2)`
- C. Both A and B
- D. None

Answer: C

Explanation: Both forms work.

Q13. What will this output?

python

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```
text = "a,b,c" print(text.split(","))
```

- A. ['a', 'b', 'c']
- B. 'a b c'
- C. ['abc']
- D. Error

Answer: A

Explanation: split(',') splits at commas.

Q14. What is output of:

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```
s = "Hello {}" print(s.format("Komal"))
```

- A. Hello {}
- B. Hello Komal
- C. Hello format
- D. Error

Answer: B

Explanation: format() replaces {} placeholders.

Q15. Which is the best way to format strings in Python 3.6+?

- A. % operator
- B. format() method
- C. f-strings
- D. join()

Answer: C

Explanation: f-strings are modern and efficient.

Q16. What is a correct way to assign values to variables a, b from list?

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lst = [1, 2]

- A. a, b = lst
- B. a = lst[0]; b = lst[1]
- C. Both A and B
- D. None

Answer: C

Explanation: Both methods unpack list elements.

Q17. How do you join list of strings words = ['I', 'am', 'Komal'] into a sentence?

- A. ''.join(words)
- B. words.join('')
- C. ''.split(words)
- D. join('', words)

Answer: A

Explanation: 'separator'.join(list) joins elements.

Q18. What happens if you omit comma in single-element tuple?

- A. It's considered a tuple

- B. It's considered as a parenthesized expression
- C. Syntax error
- D. Creates list

Answer: B

Explanation: (1) is just 1 in parentheses.

Q19. How to get length of tuple t?

- A. len(t)
- B. t.length()
- C. t.size()
- D. count(t)

Answer: A

Explanation: len() returns number of elements.

Q20. What does map() do?

- A. Filters list
- B. Applies a function to all items of an iterable
- C. Sorts list
- D. Joins list elements

Answer: B

Explanation: map() transforms each item.

🚧 Hard Level (10 Questions)

Q21. What is the difference between tuples and lists?

- A. Lists are immutable
- B. Tuples are mutable
- C. Tuples are immutable, lists mutable
- D. Both are immutable

Answer: C

Explanation: Lists can be changed, tuples cannot.

Q22. Which of these is a valid way to swap two variables in Python?

A.

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a, b = b, a

B.

python

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```
temp = a  
a = b  
b = temp
```

C. Both A and B

D. None

Answer: C

Explanation: Both methods swap variables correctly.

Q23. What does this code output?

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```
x = "5" y = int(x) + 3 print(y)
```

- A. "53"
- B. 8
- C. TypeError
- D. "8"

Answer: B

Explanation: int() converts string to number.

Q24. How does f-string differ from .format() method?

- A. f-string is evaluated at runtime, .format() is compile time
- B. f-string is more readable and concise
- C. .format() is faster
- D. None

Answer: B

Explanation: f-strings improve readability.

Q25. What will happen if you do this?

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```
a = (1, 2, 3) a[0] = 4
```

- A. Tuple changes first element
- B. Syntax error
- C. TypeError: 'tuple' object does not support item assignment
- D. No effect

Answer: C

Explanation: Tuples don't support item assignment.

Q26. What will this output?

```
python
```

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```
lst = [1, 2, 3] print([x*2 for x in lst])
```

- A. [1, 2, 3]
- B. [2, 4, 6]
- C. [1, 4, 9]
- D. Error

Answer: B

Explanation: List comprehension doubles each element.

Q27. What does this do?

python

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```
",".join(["a", "b", "c"])
```

- A. "abc"
- B. "a,b,c"
- C. "a b c"
- D. Error

Answer: B

Explanation: Joins list elements with commas.

Q28. Which of these statements about Python variables is true?

- A. Variables must be declared before use
- B. Variables are statically typed
- C. Variables store references to objects
- D. Variable names cannot be reassigned

Answer: C

Explanation: Python variables reference objects.

Q29. What is the output of this?

```
python
```

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```
a, *b = (1, 2, 3, 4) print(b)
```

A. [1, 2, 3, 4]

- B. [2, 3, 4]
- C. (2, 3, 4)
- D. Error

Answer: B

Explanation: *b collects remaining values in list.

Q30. Which of these is the correct way to format a floating-point number to 2 decimal places?

A.

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f"{pi:.2f}"

B.

python

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`"{pi:.2f}".format(pi)`

C.

python

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`format(pi, ".2f")`

D. All of the above

Answer: D

Explanation: All format float to 2 decimals.